PERIPHERAL ARTERY DISEASE

ATHEROSCLEROTIC OBSTRUCTION OF LOWER-EXTREMITY BLOOD FLOW

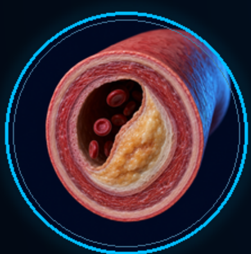
WHAT PAD IS

Peripheral artery disease is a manifestation of systemic atherosclerosis in which plaque narrows or occludes arteries supplying the limbs. Reduced flow may cause exertional leg symptoms and, in advanced disease, ischemic rest pain or tissue loss.

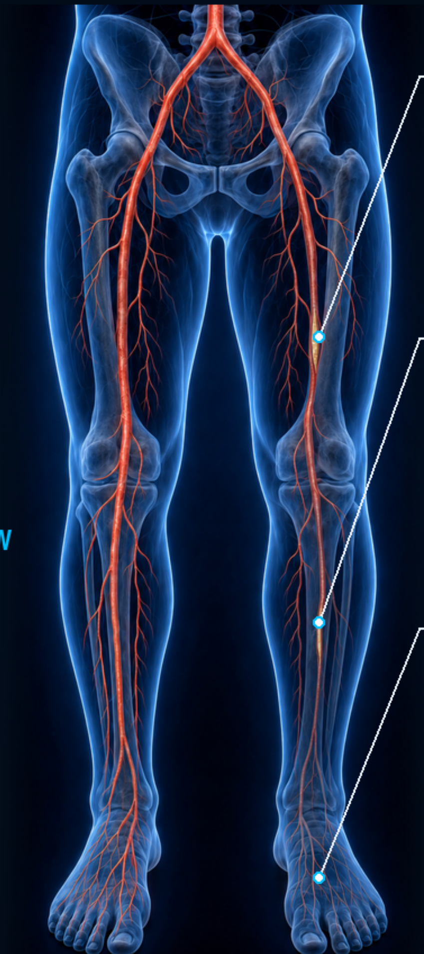
WHY IT MATTERS

PAD affects limb function and also identifies patients at elevated cardiovascular risk.

HOW PLAQUE LIMITS FLOW



Eccentric intimal plaque projects into the lumen, increasing resistance and restricting flow reserve.



FOCAL STENOSIS

Plaque narrows the superficial femoral artery and raises resistance.

REDUCED FLOW RESERVE

Resting flow may be adequate; exercise can unmask supply-demand mismatch.

DISTAL HYPOPERFUSION

Advanced disease may cause rest pain, nonhealing wounds, or gangrene.

AT A GLANCE

PAD ranges from asymptomatic disease to claudication and chronic limb-threatening ischemia. Evaluation begins with history, pulse examination, and the resting ankle-brachial index.

EDUCATIONAL PEARL

A normal-looking limb does not exclude PAD. When symptoms are exertional and the resting ABI is normal or borderline, exercise ABI testing can reveal impaired flow reserve.

IMAGE TITLE	CATEGORY	MODALITY	CLINICAL SOURCE	REVIEW
Overview of Peripheral Artery Disease	Clinical Overview	3D Medical Illustration	2024 ACC/AHA Lower-Extremity PAD Guideline	CLINICAL PENDING

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LOWER-EXTREMITY ARTERIAL ANATOMY

AORTOILIAC INFLOW • FEMOROPOPLITEAL SEGMENT • TIBIAL RUNOFF • PEDAL CIRCULATION

AORTOILIAC INFLOW

Distal abdominal aorta

Common iliac artery

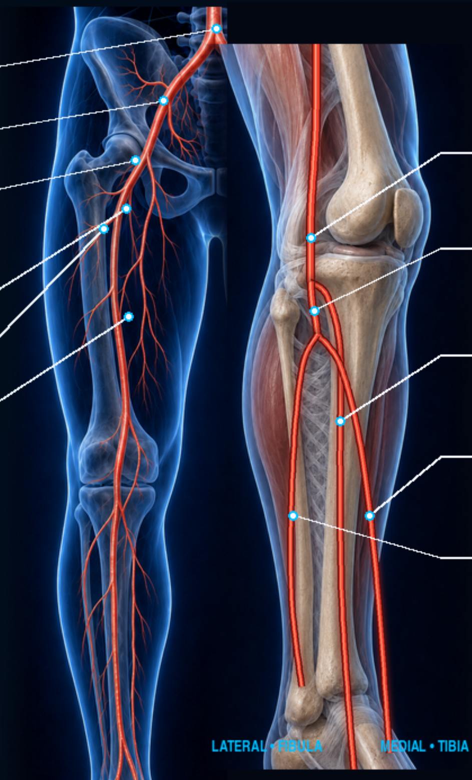
External iliac artery

FEMORAL SEGMENT

Common femoral artery

Deep femoral artery
(profunda femoris)

Femoral artery (SFA)



KNEE & TIBIAL RUNOFF

MAGNIFIED POSTERIOR-OBLIQUE VIEW

Popliteal artery

Tibiofibular trunk
(tibioperoneal trunk)

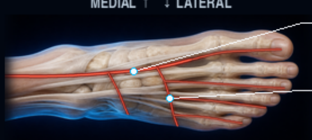
Anterior tibial artery
(to anterior compartment)

Posterior tibial artery

Fibular artery
(peroneal artery)

DORSAL FOOT • RIGHT

MEDIAL ↑ ↓ LATERAL



Dorsalis pedis artery

Arcuate artery
(variable)

ANKLE ← → TOES

PLANTAR FOOT • RIGHT

MEDIAL ↑ ↓ LATERAL

Posterior tibial artery

Medial plantar artery

Deep plantar arch

Lateral plantar artery

ANKLE ← → TOES

AT A GLANCE

The popliteal artery typically gives rise to the anterior tibial artery and a tibioperoneal trunk. The anterior tibial continues as dorsalis pedis; the posterior tibial enters the plantar foot.

EDUCATIONAL PEARL

Trace inflow to runoff: aorta → iliac → femoral → popliteal → tibial vessels → pedal circulation. Pedal branches vary, so procedural planning depends on patient-specific imaging.

IMAGE TITLE	CATEGORY	MODALITY	ANATOMY SOURCE	REVIEW
Lower-Extremity Arterial Anatomy	Anatomy	3D Medical Atlas	Cadaveric popliteal and pedal studies	CLINICAL PENDING

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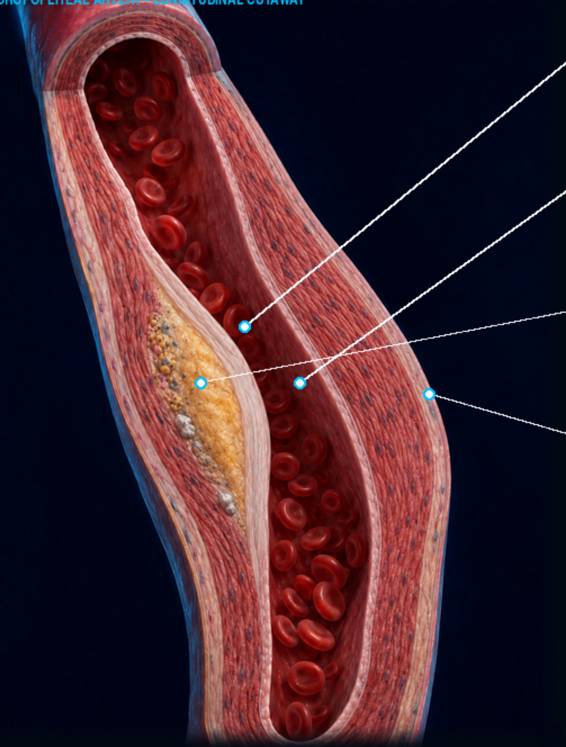
ATHEROSCLEROSIS PATHOPHYSIOLOGY

ENDOTHELIAL STRESS → LIPID RETENTION → FIBROUS PLAQUE → FLOW LIMITATION

FEMOROPOPLITEAL ARTERY • LONGITUDINAL CUTAWAY



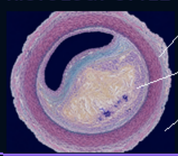
INTACT FIBROUS CAP
Collagen and smooth-muscle matrix separate blood from the necrotic core.

NARROWED LUMEN
An eccentric intimal plaque reduces pressure and exercise-flow reserve.

LIPID-NECROTIC CORE
Extracellular lipid, cholesterol debris, and foam cells expand the intima.

MUSCULAR MEDIA
The plaque is intimal; medial calcification is a distinct, often coexisting process.

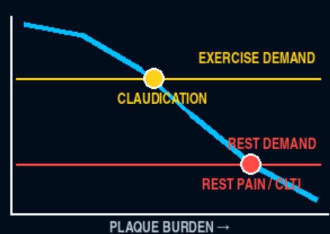
HISTOLOGY-STYLE • FIBROATHEROMA



HOW THE LESION FORMS

1. **ENDOTHELIAL STRESS** — Risk factors impair endothelial signaling
2. **LDL RETENTION** — Modified LDL is retained in the intima
3. **MONOCYTE ENTRY** — Monocytes enter and become macrophages
4. **FOAM CELLS** — Lipid-laden cells enlarge the lesion
5. **FIBROUS CAP** — Smooth muscle and collagen form the cap

WHY CLAUDICATION APPEARS



EDUCATIONAL PEARL

Claudication is demand-induced ischemia: resting flow may be adequate, but exercise hyperemia cannot overcome a flow-limiting stenosis.

IMAGE TITLE	CATEGORY	MODALITY	SOURCE	EVIDENCE
Atherosclerosis Pathophysiology	Pathophysiology	3D Illustration + Histology-Style	Rounds Codex Medical Illustration Library	High

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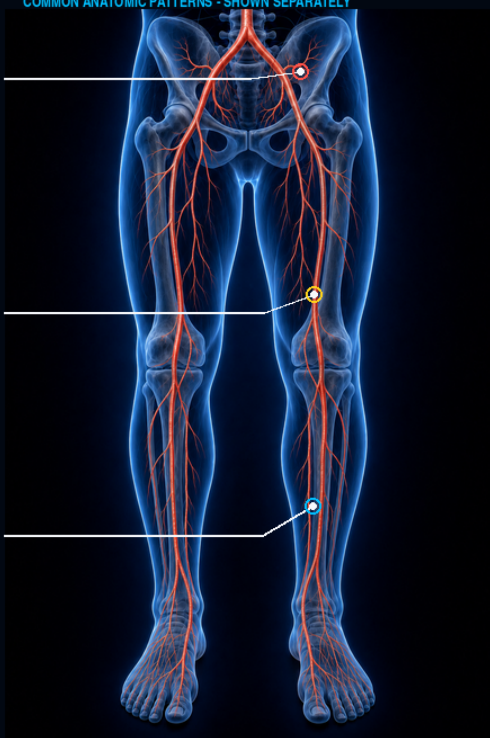
CLAUDICATION & LESION LOCALIZATION

EXERTIONAL MUSCLE ISCHEMIA OFTEN LOCALIZES DISTAL TO A FLOW-LIMITING LESION

COMMON ANATOMIC PATTERNS - SHOWN SEPARATELY



AORTOILIAC INFLOW

Buttock, hip, or thigh symptoms; calf discomfort can coexist.

FEMOROPOPLITEAL

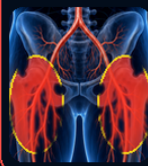
Calf claudication is the classic symptom pattern.

TIBIAL / PEDAL

Distal calf or foot symptoms often reflect multilevel or advanced PAD.

COMMON SYMPTOM REGIONS

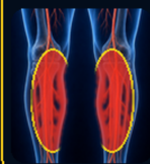
PROXIMAL INFLOW



Buttock / hip / thigh

Reduced femoral pulses may support the pattern.

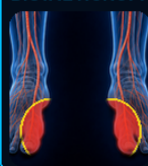
FEMOROPOPLITEAL



Calf

Femoral pulse may remain palpable; distal pulses can fall.

DISTAL RUNOFF



Foot / arch or distal calf

Isolated tibial disease more often accompanies limb threat.

SYMPTOM LOCATION SUGGESTS - BUT DOES NOT PROVE - THE LEVEL. MULTILEVEL DISEASE IS COMMON.

THE WALK-REST-WALK SIGNATURE

O2

WALK → **DEMAND RISES** → **MUSCLE PAIN** → **STAND & REST**

- **WALK** — Same workload
- **DEMAND RISES** — Flow reserve is limited
- **MUSCLE PAIN** — Cramp / ache / fatigue
- **STAND & REST** — Usually eases <10 min

RECOVERY RESETS THE EXERTIONAL THRESHOLD

VASCULAR OR NEUROGENIC?

VASCULAR	NEUROGENIC
WORKLOAD	POSITION
Standing rest helps	Sitting / flexion helps
Pulses / ABI may be abnormal	Pulses / vascular tests are usually normal

EDUCATIONAL PEARL

True vascular claudication follows workload. Relief that requires sitting or bending forward points toward spinal stenosis.

IMAGE TITLE	CATEGORY	MODALITY	SOURCE	EVIDENCE
Claudication & Lesion Localization	Clinical Localization	3D Illustration + Symptom Map	2024 ACC/AHA PAD Guideline	High

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CHRONIC & ACUTE LIMB THREAT

CLTI (>2 WEEKS) AND ALI (<2 WEEKS) REQUIRE DIFFERENT URGENCY FRAMEWORKS

CLTI - CHRONIC >2 WEEKS

ISCHEMIC REST PAIN

Forefoot pain often worsens at night or with elevation and improves with dependency.

WIfI STAGES THE LIMB THREAT

AMPUTATION RISK + EXPECTED BENEFIT OF REVASCULARIZATION

ALI - SUDDEN <2 WEEKS

THE 6 Ps

- PAIN
- PALLOR
- PULSELESSNESS
- PARESTHESIA
- PARALYSIS
- POIKILOTHERMIA

RUTHERFORD ALI VIABILITY

I	IIa	IIb	III
VIABLE	PROMPT	IMMEDIATE	IRREVERSIBLE

SENSORY LOSS OR MOTOR WEAKNESS = THREATENED LIMB

TIME & ACTION

CLTI - URGENT LIMB PRESERVATION

LIMB TEAM → **STAGE** → **RESTORE FLOW**

- **LIMB TEAM** — Vascular + wound care
- **STAGE** — Perfusion + WIfI
- **RESTORE FLOW** — Revascularize when feasible

ALI - VASCULAR EMERGENCY

ASSESS NOW → **UF HEPARIN** → **REVASCULARIZE**

- **ASSESS NOW** — Viability at bedside
- **UF HEPARIN** — Unless contraindicated
- **REVASCULARIZE** — If the limb is salvageable

EDUCATIONAL PEARL

Rest pain, tissue loss, or gangrene is not severe claudication. In ALI, sensory loss or weakness sharply increases urgency.

IMAGE TITLE	CATEGORY	MODALITY	SOURCE	EVIDENCE
Chronic & Acute Limb Threat	Limb-Threatening Ischemia	Clinical 3D + Urgency Map	2024 ACC/AHA PAD Guideline	High

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GROSS PATHOLOGY & HISTOLOGY

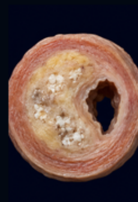
INTIMAL ATHEROMA - LUMEN LOSS - THROMBOSIS - DISTINCT MEDIAL CALCIFICATION

GROSS MORPHOLOGY

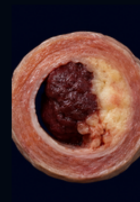
TRANSVERSE MUSCULAR ARTERY

NORMAL
Patent lumen

INTIMAL ATHEROMA
Eccentric plaque



FIBROCALCIFIC PLAQUE
Calcium within plaque

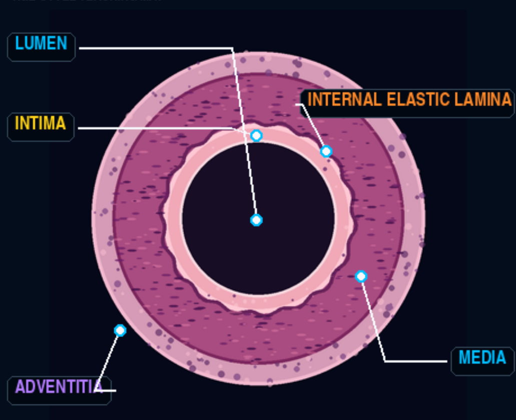


PLAQUE + THROMBUS
Superimposed luminal clot

PLAQUE IS AN INTIMAL PROCESS; THROMBUS OCCUPIES THE LUMEN

NORMAL ARTERIAL WALL

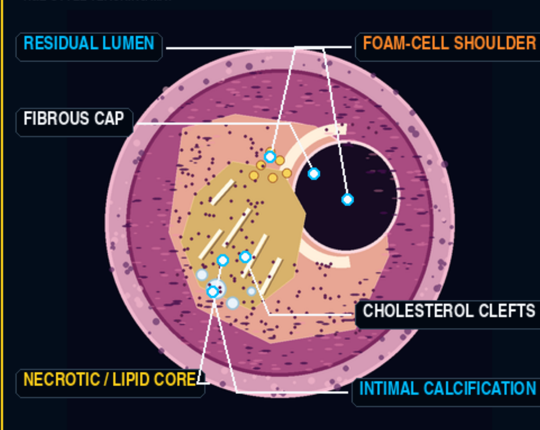
H&E-STYLE TEACHING MAP



LUMEN -> INTIMA -> MEDIA -> ADVENTITIA

ADVANCED INTIMAL FIBROATHEROMA

H&E-STYLE TEACHING MAP



MEDIA REMAINS OUTSIDE THE INTIMAL PLAQUE

MEDIAL ARTERIAL CALCIFICATION

SEPARATE FROM ECCENTRIC INTIMAL ATHEROMA

Circumferential or patchy mineral follows the medial layer.

It stiffens the artery and may preserve a visibly patent lumen.

Clinical effect: ankle pressure may be falsely high / noncompressible.

ABI
>1.40?
CHECK TBI

TEACHING ILLUSTRATIONS - NOT DIAGNOSTIC PHOTOMICROGRAPHS - PATHOLOGIST REVIEW PENDING

EDUCATIONAL PEARL

Atherosclerotic plaque is intimal. Medial arterial calcification is a distinct wall process that can stiffen vessels and make the ABI falsely elevated or noncompressible.

IMAGE TITLE	CATEGORY	MODALITY	SOURCE	VERIFICATION
Gross Pathology & Histology	Vascular Pathology	Gross + H&E-style Teaching Maps	Human peripheral artery histology	Pathology pending

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CLINICAL + PATH
PENDING

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VASCULAR EXAMINATION

LOOK - FEEL - COMPARE BILATERALLY - CONFIRM ABNORMAL FINDINGS WITH PHYSIOLOGY

POSITIONAL COLOR CHANGE - SAME LIMB

ELEVATE, THEN LOWER TO TRUE DEPENDENCY

OBSERVE COLOR CHANGE IN THE SAME LIMB

CAPILLARY REFILL - GREAT-TOE NAIL BED

DELAYED COLOR RETURN SUPPORTS HYPOPERFUSION

SKIN - HAIR - NAILS

PALPATE TEMPERATURE

COOL DISTAL SKIN + TROPHIC CHANGE MAY SUPPORT CHRONIC ISCHEMIA

CLINICAL FINDINGS - NOT DERMATOMES OR PROOF OF A SINGLE LESION LEVEL; MULTILEVEL DISEASE IS COMMON

PULSE PALPATION - EXAMINE BILATERALLY

GRADE: 0 ABSENT - 1 DIMINISHED - 2 NORMAL - 3 BOUNDING

COMMON FEMORAL

ANTERIOR GROIN

Just below the inguinal ligament.

POPLITEAL

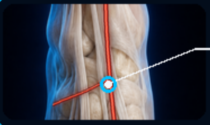
POSTERIOR-OBLIQUE KNEE

Deep in the popliteal fossa behind the knee.

POSTERIOR TIBIAL

MEDIAL ANKLE - RIGHT

Behind and below the medial malleolus.



DORSALIS PEDIS

DORSAL FOOT - RIGHT

Dorsum of foot, lateral to the EHL tendon.

EDUCATIONAL PEARL

No single examination finding proves PAD or localizes one lesion. Compare both limbs, document tissue loss, and confirm abnormal findings with ABI/TBI and waveforms.

IMAGE TITLE	CATEGORY	MODALITY	SOURCE	EVIDENCE
Vascular Examination	Clinical Examination	Clinical Photography + 3D Atlas	2024 ACC/AHA + ESVS 2024	Guideline

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DIAGNOSTIC WORKUP

PHYSIOLOGY FIRST · DEFINE ANATOMY WHEN IT WILL CHANGE MANAGEMENT

RESTING ANKLE–BRACHIAL INDEX (ABI)

SUPINE · BOTH ARMS · BOTH ANKLES

ABI = higher ipsilateral ankle SBP (in DP or PT) / higher brachial SBP (from right or left arm)

≤0.90 ABNORMAL	0.91–0.99 BORDERLINE
1.00–1.40 NORMAL	>1.40 NONCOMPRESSIBLE

Record DP and PT pressures in each ankle; use the higher ankle pressure and the higher brachial pressure.

DP = dorsalis pedis · PT = posterior tibial · SBP = systolic blood pressure

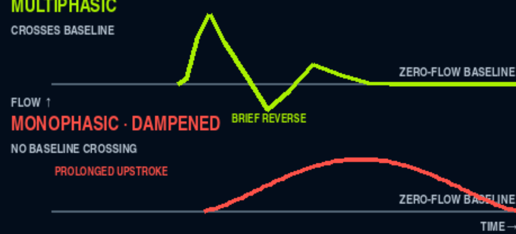
TOE PRESSURE / TBI

TBI = great-toe pressure / higher brachial SBP

≤0.70 ABNORMAL Toe pressure <30 mm Hg = severe ischemia

ARTERIAL DOPPLER WAVEFORMS

FLOW / VELOCITY vs TIME



WHAT TEST NEXT?

RESTING ABI RESULT → NEXT ACTION

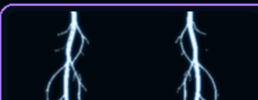
ABI ≤0.90	ABI >1.40	SYMPTOMS + ABI 0.91–1.40
PAD CONFIRMED	TBI + WAVEFORMS	EXERCISE ABI

DEFINE ANATOMY WHEN REVASCULARIZATION IS BEING CONSIDERED

NOT REQUIRED FOR EVERY PATIENT

DUPLEX
FLOW + LESION

CTA
LUMEN + CALCIUM



MRA
CONTRAST / NATIVE MRA

CATHETER DSA
PLAN / TREAT

EDUCATIONAL PEARL

A noncompressible ABI is not normal. Use toe pressure/TBI with waveforms; when exertional symptoms persist despite a normal or borderline resting ABI, perform exercise ABI.

IMAGE TITLE	CATEGORY	MODALITY	SOURCE	EVIDENCE
Diagnostic Workup	Diagnostic Testing	Clinical Photography + Medical Illustration	2024 ACC/AHA PAD + SVM/SVU Consensus	Guideline

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MANAGEMENT & REVASCULARIZATION

PREVENT CARDIOVASCULAR + LIMB EVENTS · IMPROVE WALKING · PRESERVE FUNCTION

FOUNDATION FOR EVERY PATIENT

GUIDELINE-DIRECTED MEDICAL THERAPY + LONGITUDINAL FOLLOW-UP

ANTITHROMBOTIC

Single antiplatelet for symptomatic PAD. If bleeding risk is not increased: rivaroxaban 2.5 mg BID + low-dose aspirin.

STATIN + RISK CONTROL

LDL

High-intensity statin; aim ≥50% LDL-C reduction. Treat hypertension and diabetes.

TREAT THE PERSON · PROTECT THE LIMB

STRUCTURED EXERCISE

Structured walking is core therapy. Cilostazol may improve claudication; contraindicated in heart failure.

TOBACCO + FOOT CARE

Stop tobacco with counseling + medication. Inspect and protect feet; treat wounds early.

REVASCULARIZATION — MATCH THE INDICATION, ANATOMY, AND GOALS

CLAUDICATION

Consider only if function remains limited despite GDMT + structured exercise.

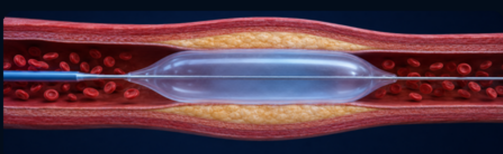
CLTI

Restore perfusion to heal wounds, relieve pain, and preserve the limb.

ALI — EMERGENCY

UFH unless contraindicated; assess viability urgently and revascularize if salvageable.

ENDOVASCULAR



Balloon angioplasty; add a stent or other device when anatomy and lesion characteristics warrant.

SURGICAL BYPASS

A conduit routes blood around an occlusion; choose target and conduit for perfusion and durability.

ENDOVASCULAR · SURGICAL · HYBRID

MULTISPECIALTY + SHARED DECISION

Choose using limb viability, lesion anatomy, available conduit, perioperative risk, expected durability, wound perfusion, and patient goals.

EDUCATIONAL PEARL

Revascularization prevents limb loss in CLTI. For claudication, use it to improve walking and quality of life only when guideline-directed therapy and structured exercise are inadequate.

IMAGE TITLE	CATEGORY	MODALITY	SOURCE	EVIDENCE
Management & Revascularization	Clinical Management	3D Illustration + Treatment Algorithm	2024 ACC/AHA PAD Guideline	Guideline

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KEY TAKEAWAYS

RECOGNIZE · CONFIRM · TREAT
PRESERVE · FOLLOW

SYSTEMIC RISK + LIMB CONSEQUENCES

CHRONIC SYMPTOMATIC

EXERTIONAL SYMPTOMS ± CLAUDICATION

CLTI

REST PAIN · ULCER · GANGRENE

ALI

SUDDEN ISCHEMIC LIMB · <2 WEEKS

1 RECOGNIZE THE PATTERN

Exertional leg symptoms relieved by rest suggest chronic symptomatic PAD. Rest pain, tissue loss, or sudden 6 Ps raise urgency.

2 EXAMINE BOTH LIMBS

Compare pulses, temperature, color, capillary refill, skin, nails, and wounds. Check footwear and protective sensation.

3 CONFIRM PHYSIOLOGY

Use resting ABI first. Add TBI with waveforms when ABI is >1.40; use exercise ABI when symptoms persist with normal or borderline ABI.

4 TREAT SYSTEMIC RISK

Use antithrombotic therapy, high-intensity statin, blood-pressure and diabetes control, tobacco cessation, structured exercise, and foot care.

5 RESTORE PERFUSION

Revascularize for CLTI to prevent limb loss; consider it for function-limiting claudication after inadequate GDMT plus structured exercise.

6 FOLLOW + ESCALATE

Monitor symptoms, function, pulses, feet, and wounds. Sudden pain, pallor, pulselessness, sensory loss, or weakness needs emergency assessment.

PATIENT ACTIONS

Take medicines as prescribed · follow the exercise plan · stop tobacco · inspect and protect both feet daily · keep follow-up

SEEK URGENT VASCULAR CARE

Sudden limb pain or coldness · new numbness or weakness · new rest pain · nonhealing wound · gangrene

EDUCATIONAL PEARL

PAD is systemic atherosclerosis with limb consequences. Confirm physiology, reduce cardiovascular and limb risk, protect the feet, restore function, and escalate limb threat without delay.

IMAGE TITLE	CATEGORY	MODALITY	SOURCE	EVIDENCE
PAD Key Takeaways	Clinical Summary	Illustrated Clinical Pathway	2024 ACC/AHA PAD Guideline	Guideline

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